Question 1

Not answered

Marked out of 1.00

Flag question

Find the output value of n if input *a* = 31.

Procedure XYZ(a: integer)

n:=0

while a  **0**

> ***n: = n + (a mod 2)***

> ***a: =***  ***\lfloor a/2 \rfloor***

Print (n)

5

Question 2

Not answered

Marked out of 1.00

Flag question

Find the output of:

Greedy Change-Making Algorithm

Input: 1100 cents

Output: The number of coins of each type: Quarters, Dimes, Nickles, Pennies

25 10 5 1

Select one:

- a. 45 , 2 , 1 , 0
- b. 43 , 2 , 0 , 5
- c. 43 , 2 , 1 , 0
- d. 44 , 0 , 0 , 0

1100 mod 25

Question 3

Not answered

Marked out of 1.00

Flag question

Find the smallest integer n such that $f(x) = O(x^n)$.

xln(x)	Choose...
$x(x^2+1)$	Choose...
4x+5	Choose...

Question 4

Not answered

Marked out of 1.00

Flag question

Given the algorithm.

procedure $f(a_1,a_2,\ldots,a_n)$

$t := 1$

$i := 1$

while (($t > 0$) and ($i < \lfloor n/2 \rfloor$))

> ***if*** $(a_i \neq a_{n-i+1})$ ***then*** $t := 0$
>
> ***else*** $i := 2i$

Let n = 200. Count the number of comparisons used in the worst case.

Question 5

Not answered

Marked out of 1.00

Flag question

A sequence of pseudorandom numbers is generated as follows

$$x_0 = 4$$

$$x_i = (6x_{i-1} + 5) \bmod 13 \text{ if } i > 0$$

Find x_4 .

Question 6

Not answered

Marked out of 1.00

Flag question

Let $m=7^5\,11^{12}\,13^{90}$ and $n = 2^4\,7^3\,11^{14}$. Choose correct statements:

(i) $\gcd(m, n) = 2^4\,13^{90}$

(ii) $\gcd(m, n) = 7^3\,11^{12}$

(iii) $\text{lcm}(m, n) = 2^4\,7^5\,11^{14}\,13^{90}$

Select one:

- a. (ii).
- b. None of the other choices is correct
- c. (ii), (iii).
- d. (i), (ii).
- e. (i), (ii), (iii).

Question 7

Not answered

Marked out of 1.00

Flag question

How many divisions are needed when using Euclidean algorithm to find the greatest common divisor of *a = 1982 and b = 1872*?

Question 8

Not answered

Marked out of 1.00

Flag question

Find octal expansion of $(B5D)_{16}$.

Answer:

Question 9

Not answered

Marked out of 1.00

Flag question

Rearrange the steps in the correct order of a proof by induction of the proposition

$1 + 3 + 5 + \ldots + (2n - 1) = n^2$ for n positive integer.

Assume that the proposition is true for $n = k$. Choos

The proposition is true for $n = 1$ because $1 = 1^2$. Choos

Then the proposition is also true for $n = k + 1$ because: $1 + 3 + 5 + \ldots + (2(k+1) - 1) = 1 + 3 + 5 + \ldots + (2k - 1) + (2k+1) = k^2 + 2k + 1 = (k+1)^2$ Choos

By induction, the proposition is true for all n positive integer. Choos

Question
10

Not answered

Marked out of 1.00

Flag question

Find a recursive definition for

$$S(n)=1^2+2^2+...+n^2.$$

Select one:

a. $S(1)=1; S(n+1)=S(n)+n^2$ for $n \geq 1$

b. $S(1)=1; S(n+1)=S(n)+2n+1$ for $n \geq 1$

c. $S(1)=1; S(n+1)=S(n)+(n+1)^2$ for $n \geq 1$

d. None of the other choices is correct

Question
11

Not answered

Marked out of 1.00

Flag question

Given the sequence

$a_0 = 1, a_1 = 2, a_n = a_{n-1} + 2a_{n-2}$ *for* $n \geq 2$

Find a_7 .

Question
12

Not answered

Marked out of 1.00

Flag question

Given the recursive algorithm.

Procedure pow(a, n: positive integers)

If n = 0 then

> ***pow(a,n):= 1***

else

> ***pow(a,n):= a.pow(a,n-1)***

Print(pow(a,n))

Find output if input = (3, 4).

Question
13

Not answered

Marked out of 1.00

Flag question

How many positive integers not exceeding 1000 and are divisible by 2 or 5?

Answer:

Question
14

Not answered

Marked out of 1.00

Flag question

A young pair of rabbits (one of each sex) is placed on an island. A pair of rabbits does not breed until they are 2 month old. After they are 2 month old they will produce 2 pairs of rabbits each month. Find the number of pairs of rabbits after 6 months.

Question
15

Not answered

Marked out of 1.00

Flag question

Given f (n)=3 f(n/2)+1, f (1)=1. Find f(8).

Answer: